

Phase 3

Advanced — Index & Study Checklist

Goal: move from applied distributed-systems patterns (Phase 2) to the deeper theory and real production-architecture concerns that separate mid-level from senior HLD answers — consensus, distributed transactions, advanced data stores, event-driven architecture, and operational resilience. Complete chapters in order; each later chapter assumes earlier Phase 1, 2, and 3 chapters.

#	Chapter	Time
1	Distributed Systems Fundamentals Fallacies of distributed computing, partial failure, clock skew	2 hr
2	Consensus Algorithms Paxos and Raft at a conceptual level, leader election, quorums	2 hr
3	Distributed Coordination Services Zookeeper and etcd, leader election, distributed locks	1.5 hr
4	Distributed Transactions Two-Phase Commit vs. the Saga pattern, compensating transactions	2 hr
5	Advanced Data Stores Cassandra, LSM trees, time-series databases	2 hr
6	OLAP vs OLTP & Data Warehousing Row vs column storage, ETL pipelines, star schema	1.5 hr
7	Batch vs Stream Processing MapReduce, Kafka Streams/Flink, windowing, Lambda architecture	2 hr
8	Event-Driven Architecture Event Sourcing, CQRS, choreography vs orchestration	2 hr
9	Real-Time Systems & Streaming Long polling, SSE, WebSockets, connection affinity at scale	2 hr
10	Bloom Filters & Probabilistic Data Structures Bloom filters, HyperLogLog, Count-Min Sketch	1.5 hr
11	Geo-Distribution & Multi-Region Design Active-active vs active-passive, GeoDNS, data residency	2 hr
12	Security at Scale AuthN vs AuthZ, sessions vs JWT, OAuth 2.0, common attack vectors	2 hr
13	Observability — Logging, Metrics, Tracing The three pillars, distributed tracing, sampling	1.5 hr

#	Chapter	Time
14	Fault Tolerance & Resilience Patterns Circuit breakers, bulkheads, retries with backoff and jitter	1.5 hr
15	Backup, Disaster Recovery & Deployment Strategies RPO/RTO, restore drills, canary/blue-green deployment	1.5 hr
16	Phase 3 Recap & Mini System Capstone: multi-region, fault-tolerant payment ledger system	2 hr

Phase 3 total: ~27 hours

Study Checklist

Mark each chapter complete only once you can do all three of the following without notes:

- Explain the core concept out loud in under 2 minutes, using the backend-engineer analogy as a bridge.
- Answer all "Common Interview Questions" from memory with a full model-approach structure.
- Complete the practice exercise(s) unaided.

Done?	Chapter
☐ Chapter 1	Distributed Systems Fundamentals
☐ Chapter 2	Consensus Algorithms
☐ Chapter 3	Distributed Coordination Services
☐ Chapter 4	Distributed Transactions
☐ Chapter 5	Advanced Data Stores
☐ Chapter 6	OLAP vs OLTP & Data Warehousing
☐ Chapter 7	Batch vs Stream Processing
☐ Chapter 8	Event-Driven Architecture
☐ Chapter 9	Real-Time Systems & Streaming
☐ Chapter 10	Bloom Filters & Probabilistic Data Structures
☐ Chapter 11	Geo-Distribution & Multi-Region Design
☐ Chapter 12	Security at Scale
☐ Chapter 13	Observability — Logging, Metrics, Tracing
☐ Chapter 14	Fault Tolerance & Resilience Patterns
☐ Chapter 15	Backup, Disaster Recovery & Deployment Strategies
☐ Chapter 16	Phase 3 Recap & Mini System

How Phase 3 Chapters Connect

Phase 3 is deliberately ordered so each chapter's mental model supports the next:

- 1 Ch 1 names the deep assumptions (partial failure, partitions, clock skew) that explain why every later chapter's technique exists.
- 2 Ch 2–4 build the coordination/correctness layer: consensus, the services built on it, and how multi-service writes stay consistent without a single database transaction.
- 3 Ch 5–7 build the advanced data layer: wide-column stores and their LSM engine, analytical (OLAP) systems, and the batch/stream processing that moves data between them.
- 4 Ch 8–9 build the event/real-time layer: event-driven architecture as a design philosophy, and the specific real-time delivery techniques it often powers.
- 5 Ch 10–11 build the efficiency/scale layer: probabilistic data structures for memory efficiency, and multi-region design for geographic scale.
- 6 Ch 12–15 build the operational layer: security, observability, resilience patterns, and backup/deployment discipline — the practices that keep a live system trustworthy.
- 7 Ch 16 forces you to combine most of the above into one working design — the real test of Phase 3.

Next step

Once every chapter is checked off and Chapter 16's capstone is solved unaided, you're ready to approve Phase 4 (Interview-Ready) — the repeatable HLD interview framework, India/UAE/Saudi panel expectations, and a full set of worked mock-interview design problems.